

Homework 11: Longitudinal Data

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Problem 27: AR(1) model (2 points)

Deriving the log-likelihood

We are given a first-order autoregressive process AR(1) with the conditional distribution:

$$P(X^t = x^t \mid X^{t-1} = x^{t-1}) = \mathcal{N}(x^t \mid Ax^{t-1}, \Gamma),$$

where $X^t \in \mathbb{R}^n$, $A \in \mathbb{R}^{n \times n}$, and $\Gamma = I_n$ (the $n \times n$ identity matrix).

The density of a multivariate normal $\mathcal{N}(x \mid \mu, \Sigma)$ is:

$$p(x \mid \mu, \Sigma) = (2\pi)^{-n/2} |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right).$$

Since $\Gamma = I_n$, we have $|\Gamma| = 1$ and $\Gamma^{-1} = I_n$. Hence each conditional density simplifies to:

$$p(x^t \mid x^{t-1}, A) = (2\pi)^{-n/2} \exp\left(-\frac{1}{2}(x^t - Ax^{t-1})^\top (x^t - Ax^{t-1})\right).$$

By the Markov property of the AR(1) process, the joint likelihood of the observations $x^1, \dots, x^T$ (conditioned on x^0) factorizes as:

$$L(A) = \prod_{t=1}^T p(x^t \mid x^{t-1}, A).$$

Taking the logarithm:

$$\ell(A) = \log L(A) = \sum_{t=1}^T \log p(x^t \mid x^{t-1}, A).$$

Substituting the Gaussian density:

$$\ell(A) = \sum_{t=1}^T \left[-\frac{n}{2} \log(2\pi) - \frac{1}{2} (x^t - Ax^{t-1})^\top (x^t - Ax^{t-1}) \right].$$

Dropping the constant term $-\frac{nT}{2} \log(2\pi)$ (which does not depend on A), the log-likelihood as a function of A is:

$$\boxed{\ell(A) = -\frac{1}{2} \sum_{t=1}^T (x^t - Ax^{t-1})^\top (x^t - Ax^{t-1}).}$$

Deriving the score equation and $\hat{A}_{\text{MLE}}$

We need to differentiate $\ell(A)$ with respect to A . Using the trace identity from the hint, for a vector $u \in \mathbb{R}^n$:

$$u^\top u = \text{tr}(u^\top u) = \text{tr}(u u^\top).$$

Let $u^t = x^t - Ax^{t-1}$. Then:

$$\ell(A) = -\frac{1}{2} \sum_{t=1}^T \text{tr}((x^t - Ax^{t-1})(x^t - Ax^{t-1})^\top).$$

Expanding the product inside the trace:

$$(x^t - Ax^{t-1})(x^t - Ax^{t-1})^\top = x^t x^{t\top} - Ax^{t-1} x^{t\top} - x^t x^{t-1\top} A^\top + Ax^{t-1} x^{t-1\top} A^\top.$$

To compute the matrix derivative $\frac{\partial \ell}{\partial A}$, we use the standard matrix calculus identities:

- $\frac{\partial}{\partial A} \text{tr}(AB) = B^\top$
- $\frac{\partial}{\partial A} \text{tr}(ABA^\top) = A(B + B^\top)$

Therefore:

$$\begin{aligned}
\frac{\partial \ell}{\partial A} &= -\frac{1}{2} \sum_{t=1}^T \left[-x^t x^{t-1\top} - x^t x^{t-1\top} + A \left(x^{t-1} x^{t-1\top} + x^{t-1} x^{t-1\top} \right) \right] \\
&= -\frac{1}{2} \sum_{t=1}^T \left[-2 x^t x^{t-1\top} + 2 A x^{t-1} x^{t-1\top} \right] \\
&= \sum_{t=1}^T \left[x^t x^{t-1\top} - A x^{t-1} x^{t-1\top} \right].
\end{aligned}$$

Score equation. Setting the score $\frac{\partial \ell}{\partial A} = 0$:

$$\sum_{t=1}^T x^t x^{t-1\top} = A \sum_{t=1}^T x^{t-1} x^{t-1\top}.$$

MLE. Solving for A :

$$\boxed{\hat{A}_{\text{MLE}} = \left(\sum_{t=1}^T x^t x^{t-1\top} \right) \left(\sum_{t=1}^T x^{t-1} x^{t-1\top} \right)^{-1}}.$$

This is the matrix analogue of the ordinary least squares estimator: we regress the “response” x^t on the “predictor” x^{t-1} , and the MLE is $\hat{A} = S_{xy} S_{xx}^{-1}$, where $S_{xy} = \sum_t x^t x^{t-1\top}$ is the cross-covariance matrix and $S_{xx} = \sum_t x^{t-1} x^{t-1\top}$ is the auto-covariance matrix of the lagged observations.

Problem 28: Modeling overdispersion in RNA-seq gene expression (8 points)

Part 1 (1 point)

We are given the hierarchical model:

$$Y_i \mid Z_i = z \sim \text{Pois}(z), \quad Z_i \sim \text{Gamma}(\alpha_i, \nu_i),$$

with $\mathbb{E}(Z_i) = \mu_i = \alpha_i \nu_i$ and $\text{Var}(Z_i) = \alpha_i^2 \nu_i$.

Marginal mean

By the **Law of Total Expectation**:

$$\mathbb{E}(Y_i) = \mathbb{E}[\mathbb{E}(Y_i \mid Z_i)].$$

Since $Y_i \mid Z_i \sim \text{Pois}(Z_i)$, we have $\mathbb{E}(Y_i \mid Z_i) = Z_i$. Therefore:

$$\boxed{\mathbb{E}(Y_i) = \mathbb{E}(Z_i) = \mu_i = \alpha_i \nu_i.}$$

Marginal variance

By the **Law of Total Variance**:

$$\text{Var}(Y_i) = \mathbb{E}[\text{Var}(Y_i \mid Z_i)] + \text{Var}(\mathbb{E}(Y_i \mid Z_i)).$$

For a Poisson random variable, $\text{Var}(Y_i \mid Z_i) = Z_i$ and $\mathbb{E}(Y_i \mid Z_i) = Z_i$. Substituting:

$$\text{Var}(Y_i) = \mathbb{E}(Z_i) + \text{Var}(Z_i) = \alpha_i \nu_i + \alpha_i^2 \nu_i.$$

$$\boxed{\text{Var}(Y_i) = \alpha_i \nu_i + \alpha_i^2 \nu_i.}$$

Since $\alpha_i^2 \nu_i > 0$, we have $\text{Var}(Y_i) = \alpha_i \nu_i + \alpha_i^2 \nu_i > \alpha_i \nu_i = \mathbb{E}(Y_i)$, confirming **overdispersion**.

Interpretation of the two terms

- $\alpha_i \nu_i = \mathbb{E}(Z_i)$ corresponds to **technical noise**. This term equals $\mathbb{E}[\text{Var}(Y_i | Z_i)]$, i.e., the average Poisson variability given a fixed expression rate. It captures the stochasticity inherent in the sequencing process (amplification, reverse transcription, library preparation), which is modeled by the Poisson layer.
- $\alpha_i^2 \nu_i = \text{Var}(Z_i)$ corresponds to **biological variability**. This term equals $\text{Var}(\mathbb{E}[Y_i | Z_i])$, i.e., the variance of the latent expression rate across samples. It reflects the sample-to-sample differences in true gene expression levels, which inflate the total variance beyond what a pure Poisson model would predict.

Part 2 (1 point)

Model A assumes a shared known shape parameter $\nu_i = \nu$ for all samples. Since $\alpha_i \nu_i = \mu_i$, we have $\alpha_i = \mu_i / \nu$.

By the hint, marginalizing over Z_i yields the negative binomial distribution:

$$P(Y_i = y) = \binom{y + \nu - 1}{y} \left(\frac{\mu_i}{\nu + \mu_i} \right)^y \left(\frac{\nu}{\nu + \mu_i} \right)^\nu.$$

Taking the logarithm:

$$\log P(Y_i = y) = \log \binom{y + \nu - 1}{y} + y \log \left(\frac{\mu_i}{\nu + \mu_i} \right) + \nu \log \left(\frac{\nu}{\nu + \mu_i} \right).$$

Identifying the EDF form

We define the **canonical parameter**:

$$\theta = \log \left(\frac{\mu_i}{\nu + \mu_i} \right).$$

From this, $e^\theta = \frac{\mu_i}{\nu + \mu_i}$ and $1 - e^\theta = \frac{\nu}{\nu + \mu_i}$. Solving for μ_i :

$$\mu_i = \frac{\nu e^\theta}{1 - e^\theta}.$$

Substituting into the log-PMF:

$$\log P(Y_i = y) = y\theta + \nu \log(1 - e^\theta) + \log \binom{y + \nu - 1}{y}.$$

This can be written in the EDF form $\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)$ with:

$$\phi = 1, \quad b(\theta) = -\nu \log(1 - e^\theta), \quad c(y, \phi) = \log \binom{y + \nu - 1}{y}.$$

Note that $c(y, \phi)$ depends only on y and ν (which is a known constant, not a function of μ_i), so it is independent of the mean parameter, confirming the EDF form.

Deriving the variance function

The mean is:

$$\mu = b'(\theta) = -\nu \cdot \frac{-e^\theta}{1 - e^\theta} = \frac{\nu e^\theta}{1 - e^\theta}. \quad \checkmark$$

The variance function is:

$$V(\mu) = b''(\theta) = \nu \cdot \frac{d}{d\theta} \left(\frac{e^\theta}{1 - e^\theta} \right) = \nu \cdot \frac{e^\theta}{(1 - e^\theta)^2}.$$

Substituting $e^\theta = \frac{\mu}{\nu + \mu}$ and $1 - e^\theta = \frac{\nu}{\nu + \mu}$:

$$V(\mu) = \nu \cdot \frac{\mu/(\nu + \mu)}{\nu^2/(\nu + \mu)^2} = \nu \cdot \frac{\mu(\nu + \mu)}{\nu^2} = \frac{\mu(\nu + \mu)}{\nu}.$$

$$\boxed{V(\mu) = \mu + \frac{\mu^2}{\nu}},$$

which is **quadratic in μ** . The total variance is $\text{Var}(Y_i) = \phi \cdot V(\mu) = \mu + \mu^2/\nu$, consistent with Part 1 under $\nu_i = \nu$ and $\alpha_i = \mu_i/\nu$.

Part 3 (3 points)

Model B assumes a shared known scale parameter $\alpha_i = \alpha$ for all samples. Since $\alpha_i \nu_i = \mu_i$, we have $\nu_i = \mu_i/\alpha$.

Showing $V(\mu) = \mu$ **with** $\phi = 1 + \alpha$

From Part 1, the marginal variance is:

$$\text{Var}(Y_i) = \alpha_i \nu_i + \alpha_i^2 \nu_i.$$

Substituting $\alpha_i = \alpha$ and $\nu_i = \mu_i/\alpha$:

$$\text{Var}(Y_i) = \alpha \cdot \frac{\mu_i}{\alpha} + \alpha^2 \cdot \frac{\mu_i}{\alpha} = \mu_i + \alpha \mu_i = (1 + \alpha) \mu_i.$$

Writing this in the form $\text{Var}(Y_i) = \phi \cdot V(\mu)$, we identify:

$$\boxed{V(\mu) = \mu, \quad \phi = 1 + \alpha,}$$

so the total variance $\text{Var}(Y_i) = (1 + \alpha)\mu$ is **linear in** μ .

Showing Y_i **does not belong to the EDF**

Under Model B, the negative binomial PMF with $\alpha_i = \alpha$ and $\nu_i = \mu_i/\alpha$ is:

$$P(Y_i = y) = \binom{y + \mu_i/\alpha - 1}{y} \left(\frac{\mu_i}{\mu_i/\alpha + \mu_i} \right)^y \left(\frac{\mu_i/\alpha}{\mu_i/\alpha + \mu_i} \right)^{\mu_i/\alpha}.$$

Simplifying the probability terms using $\mu_i/\alpha + \mu_i = \mu_i(1 + \alpha)/\alpha$:

$$\frac{\mu_i}{\mu_i(1 + \alpha)/\alpha} = \frac{\alpha}{1 + \alpha}, \quad \frac{\mu_i/\alpha}{\mu_i(1 + \alpha)/\alpha} = \frac{1}{1 + \alpha}.$$

Therefore:

$$\log P(Y_i = y) = \log \binom{y + \mu_i/\alpha - 1}{y} + y \log \left(\frac{\alpha}{1 + \alpha} \right) + \frac{\mu_i}{\alpha} \log \left(\frac{1}{1 + \alpha} \right).$$

For this to be in the EDF form $\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)$, the function $c(y, \phi)$ must be independent of μ_i (and hence of θ). However, the binomial coefficient $\binom{y + \mu_i/\alpha - 1}{y}$ depends on both y and μ_i . Expanding via the Gamma function:

$$\log \binom{y + \mu_i/\alpha - 1}{y} = \log \Gamma \left(y + \frac{\mu_i}{\alpha} \right) - \log \Gamma(y + 1) - \log \Gamma \left(\frac{\mu_i}{\alpha} \right).$$

The term $\log \Gamma(y + \mu_i/\alpha)$ couples y and μ_i in a nonlinear way that **cannot** be decomposed as a sum of a function of y alone and a function of μ_i alone. Therefore, there is no valid decomposition into $\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)$ with c independent of μ , and the distribution **does not belong to the exponential dispersion family**.

Note also that the second term $y \log(\alpha/(1 + \alpha))$ is a constant (independent of μ_i), meaning the canonical parameter θ would be constant — further confirming that the distribution cannot be parametrized in EDF form with μ as the mean.

Quasi-likelihood

Using the formula $Q_i(\mu_i; y_i) = \int_{y_i}^{\mu_i} \frac{y_i - t}{\phi \cdot V(t)} dt$ with $V(t) = t$ and $\phi = 1 + \alpha$:

$$Q_i(\mu_i; y_i) = \frac{1}{1 + \alpha} \int_{y_i}^{\mu_i} \frac{y_i - t}{t} dt = \frac{1}{1 + \alpha} \int_{y_i}^{\mu_i} \left(\frac{y_i}{t} - 1 \right) dt.$$

Evaluating:

$$Q_i(\mu_i; y_i) = \frac{1}{1 + \alpha} \left[y_i \log t - t \right]_{y_i}^{\mu_i} = \frac{1}{1 + \alpha} \left[y_i \log \mu_i - \mu_i - y_i \log y_i + y_i \right].$$

$$Q_i(\mu_i; y_i) = \frac{1}{1 + \alpha} \left[y_i \log \left(\frac{\mu_i}{y_i} \right) - (\mu_i - y_i) \right].$$

This has the same functional form as the Poisson log-likelihood (up to the scaling factor $1/(1 + \alpha)$), which is expected since the variance function $V(\mu) = \mu$ is the same as for the Poisson distribution.

Part 4 (3 points)

Now both parameters are allowed to vary with μ_i :

$$\alpha_i = \gamma + \psi \mu_i, \quad \nu_i^{-1} = \psi + \gamma \mu_i^{-1}.$$

Consistency check: $\alpha_i \nu_i = (\gamma + \psi \mu_i) \cdot \frac{1}{\psi + \gamma/\mu_i} = (\gamma + \psi \mu_i) \cdot \frac{\mu_i}{\psi \mu_i + \gamma} = \mu_i$. ✓

Special cases: When $\psi = 0$: $\alpha_i = \gamma$ (constant) and $\nu_i = \mu_i/\gamma$, recovering Model B. When $\gamma = 0$: $\alpha_i = \psi \mu_i$ and $\nu_i = 1/\psi$ (constant), recovering Model A.

Deriving the variance function

From Part 1, $\text{Var}(Y_i) = \alpha_i \nu_i + \alpha_i^2 \nu_i = \mu_i + \alpha_i^2 \nu_i$.

We compute $\alpha_i^2 \nu_i$. Since $\nu_i = \mu_i / \alpha_i = \mu_i / (\gamma + \psi \mu_i)$:

$$\alpha_i^2 \nu_i = (\gamma + \psi \mu_i)^2 \cdot \frac{\mu_i}{\gamma + \psi \mu_i} = (\gamma + \psi \mu_i) \mu_i = \gamma \mu_i + \psi \mu_i^2.$$

Therefore:

$$\text{Var}(Y_i) = \mu_i + \gamma \mu_i + \psi \mu_i^2.$$

With $\phi = 1$, the variance function is:

$$\boxed{V(\mu) = \mu + \gamma \mu + \psi \mu^2.}$$

This **interpolates** between the two special cases:

- $\psi = 0$: $V(\mu) = (1 + \gamma)\mu$, the **overdispersed Poisson** form (Model B with $\alpha = \gamma$).
- $\gamma = 0$: $V(\mu) = \mu + \psi \mu^2 = \mu + \mu^2 / \nu$, the **negative binomial** form (Model A with $\nu = 1/\psi$).

Showing Y_i does not have the EDF form

The negative binomial PMF with $\nu_i = \mu_i / (\gamma + \psi \mu_i)$ is:

$$P(Y_i = y) = \binom{y + \nu_i - 1}{y} \left(\frac{\gamma + \psi \mu_i}{1 + \gamma + \psi \mu_i} \right)^y \left(\frac{1}{1 + \gamma + \psi \mu_i} \right)^{\nu_i},$$

where we used $\mu_i / (\nu_i + \mu_i) = (\gamma + \psi \mu_i) / (1 + \gamma + \psi \mu_i)$ and $\nu_i / (\nu_i + \mu_i) = 1 / (1 + \gamma + \psi \mu_i)$.

The log-PMF is:

$$\log P(Y_i = y) = \log \binom{y + \nu_i - 1}{y} + y \log \left(\frac{\gamma + \psi \mu_i}{1 + \gamma + \psi \mu_i} \right) + \nu_i \log \left(\frac{1}{1 + \gamma + \psi \mu_i} \right).$$

Since $\nu_i = \mu_i / (\gamma + \psi \mu_i)$ depends on μ_i , the binomial coefficient $\binom{y + \nu_i - 1}{y}$ contains $\log \Gamma(y + \nu_i)$, which couples y and μ_i in a way that cannot be decomposed into $\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)$ with c independent of μ . Therefore, the distribution **does not belong to the exponential dispersion family**.

Quasi-likelihood

Using $Q_i(\mu_i; y_i) = \int_{y_i}^{\mu_i} \frac{y_i - t}{V(t)} dt$ with $V(t) = t(1 + \gamma + \psi t)$:

$$Q_i(\mu_i; y_i) = \int_{y_i}^{\mu_i} \frac{y_i - t}{t(1 + \gamma + \psi t)} dt.$$

We perform a **partial fraction decomposition**. Writing:

$$\frac{y_i - t}{t(1 + \gamma + \psi t)} = \frac{A}{t} + \frac{B}{1 + \gamma + \psi t},$$

we require $y_i - t = A(1 + \gamma + \psi t) + Bt$. Setting $t = 0$: $A = \frac{y_i}{1 + \gamma}$. Setting $1 + \gamma + \psi t = 0$: $B = -\frac{1 + \gamma + \psi y_i}{1 + \gamma}$.

Verification: $A(1 + \gamma) + (A\psi + B)t = y_i + \left(\frac{y_i\psi}{1 + \gamma} - \frac{1 + \gamma + \psi y_i}{1 + \gamma}\right)t = y_i - t$. ✓

Substituting and integrating:

$$Q_i = \frac{y_i}{1 + \gamma} \int_{y_i}^{\mu_i} \frac{dt}{t} - \frac{1 + \gamma + \psi y_i}{1 + \gamma} \int_{y_i}^{\mu_i} \frac{dt}{1 + \gamma + \psi t}.$$

The first integral is $\log(\mu_i/y_i)$. For the second, substituting $u = 1 + \gamma + \psi t$:

$$\int_{y_i}^{\mu_i} \frac{dt}{1 + \gamma + \psi t} = \frac{1}{\psi} \log\left(\frac{1 + \gamma + \psi \mu_i}{1 + \gamma + \psi y_i}\right).$$

Therefore:

$$Q_i(\mu_i; y_i) = \frac{y_i}{1 + \gamma} \log\left(\frac{\mu_i}{y_i}\right) - \frac{1 + \gamma + \psi y_i}{\psi(1 + \gamma)} \log\left(\frac{1 + \gamma + \psi \mu_i}{1 + \gamma + \psi y_i}\right).$$

Verification of special cases:

- $\psi \rightarrow 0$ (**overdispersed Poisson, Model B**): Using $\log(1 + x) \approx x$ for small x , the second term approaches $\frac{1 + \gamma}{\psi(1 + \gamma)} \cdot \frac{\psi(\mu_i - y_i)}{1 + \gamma} = \frac{\mu_i - y_i}{1 + \gamma}$, giving $Q_i = \frac{1}{1 + \gamma} [y_i \log(\mu_i/y_i) - (\mu_i - y_i)]$, consistent with Part 3.
- $\gamma = 0$ (**negative binomial, Model A**): $Q_i = y_i \log(\mu_i/y_i) - \frac{1 + \psi y_i}{\psi} \log\left(\frac{1 + \psi \mu_i}{1 + \psi y_i}\right)$, the negative binomial quasi-likelihood with $\nu = 1/\psi$.